

Functional Symbolic Bifurcation Networks

Modelling AI and LLM Interaction as a Finite Nonlinear Dynamical
Process

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Abstract

This monograph develops a finite dynamical framework for examining interactions with artificial intelligence and large language models. Rather than treating a conversation as a collection of independent prompts and answers, it treats the interaction as a Functional Symbolic Trajectory: a finite, ordered and provenance-bearing chain in which each insertion alters the condition from which the next output is generated. The model begins with a start-up network condition, proceeds through a recursive train of conditioned filter events, and permits a held trajectory to be presented to several separate models. Such presentations are not classically parallel. They constitute an approximately parallel, externally registered split in which the same visible symbolic chain is repartitioned by different model-specific filters. Once outputs are returned to their respective histories, daughter trajectories emerge and may diverge, stabilise, cross-couple or reconverge.

The resulting structure is termed a *Functional Symbolic Bifurcation Network*. It is formulated as a directed, provenance-bearing network of finite symbolic states and observed transformations. The framework distinguishes architectural splitting from classical dynamical bifurcation, external symbolic equivalence from endogenous identity, and a model's visible answer from the wider nested filtering process through which that answer is produced. It also provides a practical experimental method for replaying, branching, continuing and recombining conversational chains across different models. Possible applications include comparative model study, the measurement of meaning divergence, trajectory-sensitive control, scientific argument tracing, model-update monitoring and the development of Takens Based Transformers. Open questions concerning semantic distance, hidden state, stochasticity, branch identity, context compression and the role of the human participant are retained as part of the framework rather than prematurely closed.

Keywords: Functional Symbolic Dynamics; Functional Symbolic Trajectory; large language model; nonlinear dynamics; symbolic bifurcation; filter network; trajectory reconstruction; provenance; Takens Based Transformer; Dynamic Semantic Lattice.

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Chapter 1

Introduction: From Exchanges to Trajectories

An interaction with a large language model is ordinarily displayed as a sequence of alternating messages. A person inserts a prompt, the model returns an answer, another prompt follows, and the exchange continues. The graphical arrangement encourages a simple interpretation: each prompt is an input, each answer is an output, and the model is a stable function operating between them. This description is useful at the level of an interface, but it suppresses the developing history through which each later response becomes possible.

The second question in a conversation is not processed under the same symbolic condition as the first. It arrives after the first answer has been generated, read, accepted, rejected, qualified or redirected. When the earlier exchange is retained in the context, that exchange becomes part of the material from which the next active model condition is reconstructed. The output is therefore not merely a terminal product. Once returned to the conversation record, it becomes part of a later input condition. The apparently linear input–output arrangement closes into a recurrent symbolic process.

For the present framework, the conversation is treated as a finite observed trajectory. It has a registered beginning, an ordering, a changing extent and a provenance. Its constituents are not only the words written by the human participant. They include the visible model outputs, the system framing where this is known, the identity and version of the model, the rules by which earlier messages are retained or removed, and the conditions under which the response is decoded. Some of these constituents are directly registered; others remain inferred or unavailable. The distinction matters. Functional Symbolic Dynamics does not require complete access to the internal activations of a model. It begins from the finite transformations that can be registered and from the conditions that can be recorded.

This creates a practical opportunity. A conversational chain can be held and presented to another model. It can also be copied and presented to several models. Each model carries its own learned partition, instruction hierarchy, tokenisation process, alignment history and decoding arrangement. The held visible chain may therefore be the same at the external level while the reconstructed internal conditions are not. The result is not a classical parallel experiment in which identical systems receive an identical physical state at an identical moment. It is an approximately parallel symbolic presentation followed by

a split into separately registered continuations.

The word *bifurcation* is used here with care. A classical bifurcation is normally a qualitative change in dynamical behaviour associated with variation in a parameter. The initial split considered in this monograph is instead an architectural or functional event: one recorded trajectory is copied or routed to more than one conditioned filter. This will be called a *Functional Symbolic Bifurcation*. Classical-like changes of response regime may subsequently be investigated within the branches, but they are not assumed merely because the branches differ.

The purpose of this monograph is to construct this idea as a working model. The objective is not to redescribe language models with the vocabulary of nonlinear dynamics for rhetorical effect. It is to specify the finite observable objects, the transformations acting upon them, the branch conditions and the measurements that would make the proposal experimentally useful. The framework is consequently both conceptual and operational. It belongs within Functional Symbolic Dynamics, while also providing a possible architecture for multi-model systems and trajectory-sensitive models such as the Takens Based Transformer.

Chapter 2

Conceptual Framing

2.1 Functional Symbolic Dynamics

Functional Symbolic Dynamics begins from finite registered distinctions. A symbol is not treated as an unbounded abstract object detached from its production and use. It is a finite mark or state whose practical identity depends upon admissible distinction, provenance and context. A symbolic trajectory is an ordered pathway through such states. The pathway carries not only content but also constraint, uncertainty, historical placement and conditions of admissibility.

A Functional Symbolic Trajectory may be written provisionally as

$$\mathcal{T}_n = \left(v_0 \xrightarrow{o_1} v_1 \xrightarrow{o_2} v_2 \cdots \xrightarrow{o_n} v_n \right), \quad (2.1)$$

where each v_i is a registered symbolic state and each o_i is a finite operation or observed transition. The trajectory is not simply the ordered list $v_0, v_1, \dots, v_n$. Its edges matter. Removing the operations, ordering and conditions that connect the states compresses away the path by which their later meanings were established.

In an LLM interaction, the visible transcript offers a particularly accessible symbolic trajectory. It can be copied, compared and re-presented. Yet the transcript is not the whole physical system. Electrical transitions, software processes, network transport, stored weights and human perception all participate. Functional Symbolic Dynamics does not claim that these disappear. It establishes a disciplined boundary: the transcript is the principal externally registered signal, while the internal model state is reconstructed, partially inferred or left unspecified according to the available evidence.

2.2 The observed and the inferred

The framework distinguishes three levels. First, there is the *registered exogenous trajectory*: the messages, ordering, timestamps, model labels and known settings that can be stored. Secondly, there is the *inferred endogenous condition*: the temporary activation pattern or operative model state that is inferred to support generation but is usually not available to an external observer. Thirdly, there is the *projected continuation*: the finite output returned to the shared symbolic record.

The distinction prevents the visible text from being mistaken for an inspection of the

model’s internal meaning. It also prevents an inaccessible hidden state from becoming a necessary fictional object. The framework can remain useful when only the external chain and its visible continuations are available. Internal measurements can be added when accessible, but they are not silently presumed.

2.3 Why the term filter is useful

A model may be considered a filter in the broad functional sense that it receives a finite symbolic condition and permits, amplifies, redirects or suppresses possible continuations. This filter is not merely a final moderation step. It is a nested composition that may include input partitioning, tokenisation, contextual reconstruction, learned parameter constraints, instruction following, decoding and output qualification.

We may write the model-conditioned filter as

$$F_j = F_{\text{out}}^{(j)} \circ F_{\text{decode}}^{(j)} \circ F_{\text{trajectory}}^{(j)} \circ F_{\text{partition}}^{(j)}. \quad (2.2)$$

This expression is a functional decomposition, not a claim that every deployed model exposes or implements four literal modules with these names. Its purpose is to resist the idea that a raw, model-independent meaning is produced first and only afterwards filtered. The model’s entire construction conditions which continuations can become visible.

2.4 Nonlinearity and path dependence

The process is nonlinear in a practical sense because a symbolic insertion does not have a fixed effect independent of the state into which it is inserted. The same sentence can redirect one trajectory, stabilise another and leave a third almost unchanged. Outputs are returned to the later context, so small differences can alter subsequent reconstructions. Context limits, compression and selection can also remove earlier distinctions, producing discontinuous changes in the effective history.

The proposal does not require an immediate assertion that an LLM conversation is mathematically equivalent to a classical continuous nonlinear system. The language process is finite, partitioned, path-dependent and often stochastic. Functional Symbolic Dynamics begins from those properties. Classical tools may later be borrowed where their assumptions can be satisfied or suitably translated.

Chapter 3

The Start-Up Network Condition

3.1 No conversation begins from nothing

The phrase *start-up network condition* identifies the inherited condition from which the first visible continuation is generated. A blank conversation window is not a blank model. The model already carries a trained response landscape, a tokenisation scheme, an instruction structure, an interface and a decoding policy. The initial prompt selects and perturbs within that inherited organisation.

For model j , a provisional start-up condition can be written as

$$S_0^{(j)} = (\theta_j, \tau_j, \iota_j, \gamma_j, \delta_j, m_j, H_0), \quad (3.1)$$

where θ_j denotes the learned parameter condition, τ_j the symbolic partition or tokenizer, ι_j the instruction and identity framing, γ_j known alignment and admissibility constraints, δ_j decoding settings, m_j the model and version record, and H_0 the initial externally supplied context. In an actual study, some of these entries may be unknown. Unknown does not mean absent; it means that uncertainty must remain attached to the record.

The start-up condition is therefore both physical and symbolic, but it is not equally observable at every level. Functional Symbolic Dynamics records the available condition rather than completing it with an imagined exact state.

3.2 The initial condition as a basin selection

The learned model may be pictured as supporting many possible response basins. A system instruction, initial role, example dialogue or long run-in can make some continuations more accessible and others less admissible. It is tempting to describe this as the prompt locating a point in a latent semantic space. Within the present framework, a more cautious formulation is available: the start-up material establishes a finite observed condition from which a particular family of continuations is repeatedly registered.

The basin is consequently inferred from trajectory behaviour. If related insertions repeatedly return to a stable voice, explanatory structure or conceptual vocabulary, that regularity can be treated as evidence of a response basin. The basin is not required to be a directly visible geometric container.

3.3 The human participant as part of the start-up condition

The initial condition also includes the human participant's expectations, selections and later judgements, although these are not wholly contained in the model context. The person decides what to insert, what to ignore and which output to return to the trajectory. In an extended interaction, the human is therefore not an external spectator. The observed conversation is produced by a coupled system in which human and model alternately perturb and filter the developing chain.

This point becomes important when the same chain is presented to several models. The person may unconsciously alter punctuation, timing, framing or follow-up questions in response to each model. A rigorous experiment must decide whether the human remains an active adaptive participant or whether a previously registered insertion sequence is replayed without intervention. Both arrangements are useful, but they measure different systems.

Chapter 4

The Recursive Filter Train

4.1 A finite conversational history

Let a visible conversational history after n turns be

$$H_n = S_{\text{vis}} \parallel (u_1, y_1) \parallel (u_2, y_2) \parallel \cdots \parallel (u_n, y_n), \quad (4.1)$$

where S_{vis} is the visible part of the start-up condition, u_i is an insertion and y_i is the registered model output. The symbol $\parallel$ denotes finite ordered concatenation. It does not imply that every element is retained without compression by the implementation.

At the next turn, model j receives a model-specific presentation of the history and new insertion:

$$z_{n+1}^{(j)} = R_j(W_j(H_n \parallel u_{n+1})). \quad (4.2)$$

Here W_j is the known or inferred interface wrapping and R_j is the reconstruction of an active model condition. A projected output is then registered:

$$y_{n+1}^{(j)} = P_j(F_j(z_{n+1}^{(j)}; \eta_{n+1}^{(j)})), \quad (4.3)$$

where P_j denotes projection into a finite visible response and $\eta_{n+1}^{(j)}$ collects sampling or other contingent conditions. The history becomes

$$H_{n+1}^{(j)} = H_n \parallel (u_{n+1}, y_{n+1}^{(j)}). \quad (4.4)$$

4.2 The output becomes part of the later condition

Figure 4.1 displays the central recurrence. The visible answer is not simply removed after delivery. When it is retained in the transcript, it participates in the next reconstruction. This produces the feedback through which a conversational trajectory develops.

4.3 A train whose blocks are history-conditioned

The conversation may be drawn as a train of filter blocks:

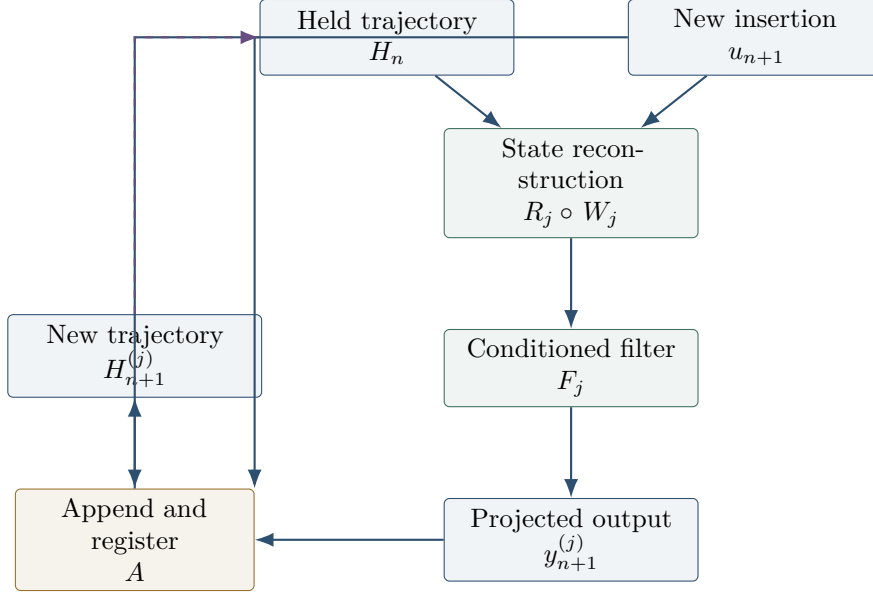


Figure 4.1: The recursive filter event. The dashed return marks the use of the newly registered history in a later turn.

$$H_0 \xrightarrow{u_1, F_j^{[1]}} H_1 \xrightarrow{u_2, F_j^{[2]}} H_2 \xrightarrow{u_3, F_j^{[3]}} \dots \quad (4.5)$$

The superscript $[n]$ does not necessarily indicate a change in the stored model parameters. It records that the effective action of the filter is conditioned by a different history:

$$F_j^{[n]}(\cdot) = F_j(\cdot \mid H_{n-1}). \quad (4.6)$$

Thus the filter train is not a row of independent identical operations. The later blocks inherit the registered consequences of earlier blocks. The diagram is linear as a representation, but the function is recursive and path-dependent.

4.4 Externalised memory and reconstructed state

In many practical LLM arrangements, temporary activation states do not persist unchanged between separate inference calls. The conversational record is re-presented and an operative state is reconstructed from it. Precise implementations vary, and some services may retain additional state that is not visible to the user. The general observation remains useful: the transcript functions as an external symbolic memory through which continuity can be reconstructed.

This gives the interaction a Takens-like character without requiring the claim that ordinary attention is literally a classical delay embedding. A finite observed sequence carries enough ordered information for a model to reconstruct a condition that supports continuation. The reconstructed condition is model-relative, and the visible continuation provides a new measurement appended to the sequence.

Chapter 5

Approximate Parallelism and the Functional Split

5.1 Why the branches are not classically parallel

Suppose a held trajectory H_n is presented to models A , B and C . At the interface, the text can be made exactly identical at the byte or character level. Yet the physical and computational presentations differ. They occur through different tokenisers, learned parameters, instruction hierarchies, machines, service conditions and perhaps different times. Even where the same model family is used, version changes, sampling and hidden service conditions can disturb equivalence.

We can therefore state only an external symbolic equivalence:

$$H_n^{(A)} \text{extsim} H_n^{(B)} \text{extsim} H_n^{(C)}. \quad (5.1)$$

The corresponding internal reconstructions need not be equal:

$$R_A(W_A(H_n)) \neq R_B(W_B(H_n)) \neq R_C(W_C(H_n)). \quad (5.2)$$

The approximate parallelism is relational rather than strictly simultaneous. The branches share a declared parent record. They do not share one endogenous state.

5.2 Definition of Functional Symbolic Bifurcation

Definition. A *Functional Symbolic Bifurcation* occurs when one held finite symbolic trajectory is copied, routed or re-presented to two or more separately conditioned filters, producing distinct registered continuations whose common parentage is preserved by provenance.

Let B_n be a registered branch event. Then

$$B_n(H_n) = \{W_A(H_n), W_B(H_n), \dots, W_J(H_n)\}. \quad (5.3)$$

Each branch yields

$$y_{n+1}^{(j)} = P_j(F_j(R_j(W_j(H_n)))) , \quad (5.4)$$

and therefore

$$H_{n+1}^{(j)} = H_n \parallel y_{n+1}^{(j)}. \quad (5.5)$$

Figure 5.1 shows the parent trajectory and its daughter continuations.

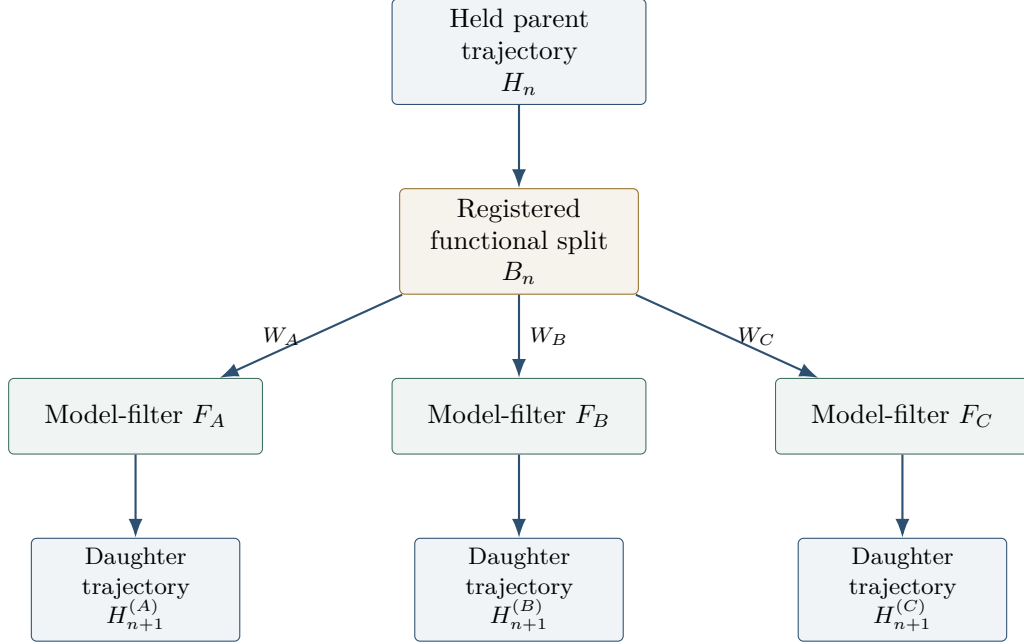


Figure 5.1: A Functional Symbolic Bifurcation. The common parent is externally registered; each filter reconstructs a distinct operative condition.

5.3 The daughter trajectories

At the split, the presentations can be treated as externally equivalent. After the first outputs are returned, the branches acquire different histories:

$$H_{n+1}^{(A)} \not\sim_{\text{ext}} H_{n+1}^{(B)} \not\sim_{\text{ext}} H_{n+1}^{(C)}. \quad (5.6)$$

If an identical new string u_{n+2} is supplied to every branch, it is not dynamically identical because it enters different histories:

$$y_{n+2}^{(A)} = \Phi_A(H_{n+1}^{(A)}, u_{n+2}), \quad (5.7)$$

$$y_{n+2}^{(B)} = \Phi_B(H_{n+1}^{(B)}, u_{n+2}), \quad (5.8)$$

$$y_{n+2}^{(C)} = \Phi_C(H_{n+1}^{(C)}, u_{n+2}). \quad (5.9)$$

This produces an important principle:

Symbolic identity of an insertion does not imply trajectory identity of its function.

The function of a sentence depends upon the branch into which it is inserted. A qualification may stabilise one branch and disrupt another. A repeated question may cause one model to revise its position while causing another to reinforce it. Meaning is therefore examined through the changing pathway, rather than assigned solely to the isolated string.

Chapter 6

Architectural Splitting and Dynamical Bifurcation

6.1 Two meanings that should not be collapsed

The architectural split described above does not by itself establish a classical dynamical bifurcation. Different models may return different wording while remaining in functionally similar response regimes. Conversely, a single model may undergo a qualitative change in response following a small symbolic perturbation. The framework therefore distinguishes:

1. **Architectural or functional splitting:** one held trajectory is deliberately routed to more than one model-conditioned filter.
2. **Behavioural regime transition:** a finite change in an insertion or condition produces a marked alteration in the structure of continuation.

The first provides an experimental apparatus for studying the second. It allows nearly matched parent trajectories to be presented to the same or different filters while their resulting paths are retained.

6.2 A finite symbolic perturbation experiment

Consider two histories that differ by one qualification:

$$H'_n = H_n \parallel \text{“Perhaps this follows.”}, \quad (6.1)$$

$$H''_n = H_n \parallel \text{“Therefore this follows.”}. \quad (6.2)$$

The visible difference is finite and recordable. If subsequent responses from the same model remain structurally similar, the perturbation may have produced only local variation. If one branch preserves an open conjectural frame while the other moves into settled exposition and continues to suppress uncertainty across later turns, a response-regime transition may have occurred.

The observation remains finite. It does not require the construction of an inaccessible continuous parameter. Repeated perturbations can nevertheless be ordered, for example by

varying modal force, qualification, role instruction or the number of supporting examples. A family of trajectories can then reveal regions of stability and transition.

6.3 Stochastic variation is not automatically bifurcation

Repeated runs from the same recorded parent may return different outputs because decoding is stochastic or because hidden implementation conditions have changed. Variation alone is insufficient evidence of a distinct response basin. A useful behavioural bifurcation should have persistence: the altered response structure should influence later turns, recur across controlled repetitions or produce a measurable change in the admissible continuation pattern.

This suggests that the unit of observation should not be one answer. It should be a continuation segment of sufficient length to test whether a distinction remains active. The appropriate length is itself an empirical question and may vary with the phenomenon being examined.

Chapter 7

The Functional Symbolic Bifurcation Network

7.1 Network definition

A Functional Symbolic Bifurcation Network can be represented as

$$\mathcal{G} = (V, E, \Lambda, \Pi), \quad (7.1)$$

where V is a finite set of registered trajectory states, E is a finite set of observed transitions, Λ labels each transition with its insertion and operative filter, and Π carries provenance and uncertainty.

A node may be written as

$$v_i = (H_i, m_i, s_i, t_i, p_i), \quad (7.2)$$

where H_i is the held symbolic chain, m_i the identified model or participant, s_i the known settings, t_i the registered time and p_i the provenance record. An edge is

$$e_i : v_i \xrightarrow{(u_i, F_j)} v_{i+1}. \quad (7.3)$$

A split node has more than one registered outgoing transition:

$$\deg^+(v_i) > 1. \quad (7.4)$$

A re-entry or synthesis node receives material from more than one earlier branch. It does not reverse the split. It creates a new trajectory with compound provenance.

7.2 The Dynamic Semantic Lattice

If branch identity j is placed on one axis and trajectory position n on another, the network forms a finite Dynamic Semantic Lattice. A lattice node is the registered interaction between a held history and a particular filter:

$$N_{j,n} = (F_j, H_n, y_n^{(j)}, p_n^{(j)}). \quad (7.5)$$

The vertical direction in Figure 7.1 follows the development of a branch. The horizontal direction compares how different filters act at a declared common trajectory position. The term lattice does not imply an infinitely extended or uniformly spaced mathematical grid. It identifies a finite organised set of registered nodes and relations.

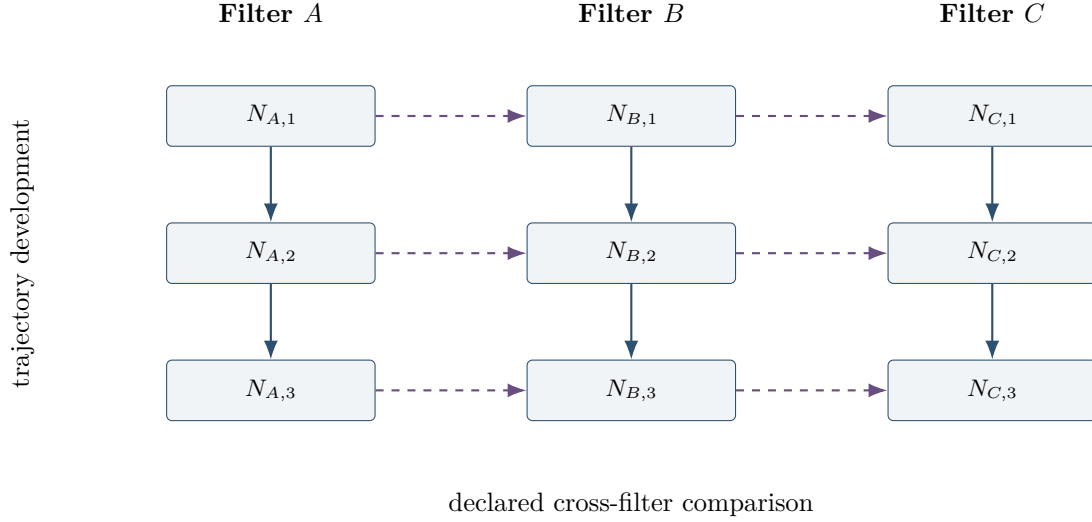


Figure 7.1: A finite Dynamic Semantic Lattice. Solid arrows follow branch development; dashed arrows mark cross-filter comparison.

7.3 A Nodal Value Function

To analyse the lattice, a node may be assigned a finite value record rather than one supposedly complete scalar score:

$$\nu(N_{j,n}) = (r_{j,n}, q_{j,n}, h_{j,n}, a_{j,n}, d_{j,n}), \quad (7.6)$$

where r records retained distinctions, q registered uncertainty or qualification, h provenance continuity, a admissibility status and d directional continuity with the parent trajectory. These components are provisional. Their operational definition must be matched to the task. In some studies they may be binary or ordinal human registrations; in others they may be computed from a labelled corpus or a trajectory-sensitive model.

The Nodal Value Function does not claim to reduce meaning to one number. It provides a way of attaching measurable finite characteristics to a node while preserving the underlying trajectory and provenance.

7.4 Branch, crossing and convergence

Three network events should be kept distinct. A *branch* occurs when a parent trajectory produces more than one daughter continuation. A *crossing* occurs when content from one branch is inserted into another. A *convergence* occurs when two or more branches produce functionally similar continuations or are combined into a new synthesis.

Apparent convergence does not prove restoration of an original common meaning. Models may converge because they share training sources, alignment conventions or dominant public vocabulary. Equally, branches may converge at the level of wording while retaining different functional histories. A convergence measure must therefore examine what later continuations remain possible, not only surface similarity at the meeting point.

Chapter 8

Three Experimental Arrangements

8.1 Common-parent replay

In common-parent replay, one held chain is presented separately to several filters and each immediate continuation is collected:

$$H_n \longrightarrow \left\{ y_{n+1}^{(A)}, y_{n+1}^{(B)}, y_{n+1}^{(C)} \right\}. \quad (8.1)$$

This is the cleanest arrangement for local filter comparison. It asks how several models reconstruct and continue one declared parent. The outputs should not be interpreted as errors distributed around a hidden ideal answer. They are model-conditioned projections of the same external symbolic record.

Common-parent replay can also be performed repeatedly with the same model. That reveals within-filter variation and gives a baseline against which cross-filter differences can be considered. A model comparison without repeated runs may confuse ordinary sampling variation with a stable difference between filter landscapes.

8.2 Independent branch continuation

In independent branch continuation, the first output from each filter remains within its daughter history. A common sequence of later insertions may then be added to all branches:

$$H_n^{(j)} \xrightarrow{u_{n+1}} H_{n+1}^{(j)} \xrightarrow{u_{n+2}} H_{n+2}^{(j)} \longrightarrow \dots. \quad (8.2)$$

This arrangement measures accumulated divergence. The later insertion strings may remain identical, but their functions become branch-relative. It is therefore suitable for investigating trajectory memory, path dependence, the persistence of conceptual distinctions and eventual reconvergence.

8.3 Sequential model cascade

In a cascade, the output from one model becomes material for the next:

$$H_n \xrightarrow{F_A} y_A \xrightarrow{F_B} y_{AB} \xrightarrow{F_C} y_{ABC}. \quad (8.3)$$

This is a true train of different filters rather than an approximately parallel split. It measures accumulated transformation. One can ask which distinctions survive repeated rewriting, whether uncertainty is removed, whether provenance is compressed, and whether unusual language is gradually returned to a dominant consensus vocabulary.

Figure 8.1 contrasts the three arrangements.

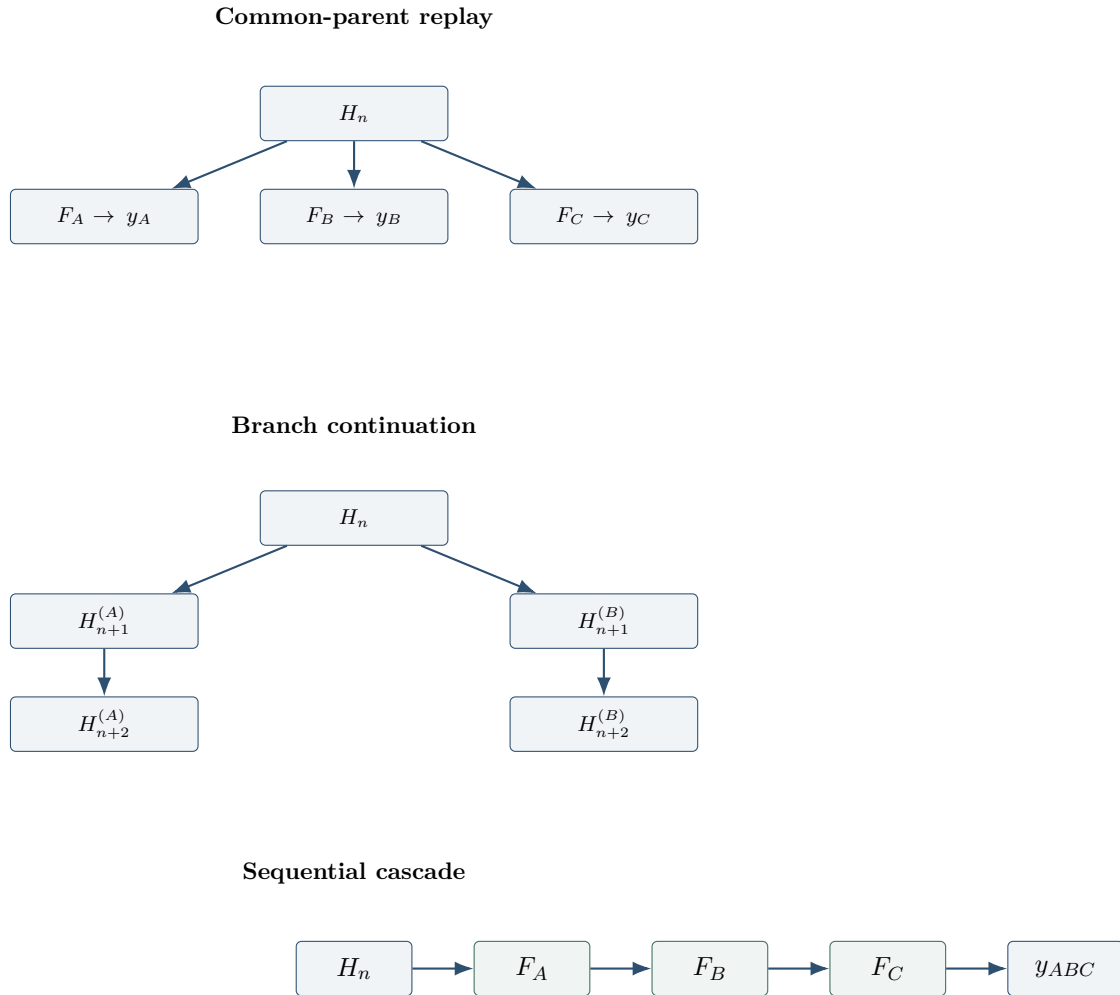


Figure 8.1: Three distinct experimental arrangements. They should not be combined under a single use of the word “parallel”.

Chapter 9

A Practical Experimental Framework

9.1 The experimental object

The basic experimental object is not an isolated prompt. It is a *trajectory package*: a finite conversational record together with its branch point, presentation rules and provenance. The package should contain enough information for another investigator, or the same investigator at a later date, to reconstruct the declared external condition.

At minimum, the package should record the parent chain, turn boundaries, speaker labels, branch insertion, date and time, model and version as displayed, known system or role instructions, known decoding settings, interface used, and whether the model retained any additional memory. Unknown settings should be explicitly marked unknown rather than omitted.

9.2 A four-stage protocol

The working protocol has four stages.

9.2.1 Stage one: register the parent

The parent trajectory is stored in a canonical form. Whitespace, message boundaries and speaker identities are retained. A checksum may be added to show that nominally common-parent presentations arose from the same stored record. The checksum establishes file identity, not semantic or physical identity within the receiving models.

9.2.2 Stage two: declare the split

The branch event is assigned an identifier. Each target model receives a declared copy of the parent and the same branch insertion where common-parent comparison is intended. Any model-specific wrapping that can be observed is recorded. Execution times are retained because the runs are approximately parallel, not necessarily simultaneous.

9.2.3 Stage three: continue or hold the branches

For a local comparison, each first output is stored and the branch is stopped. For a divergence study, every output remains in its own branch and a pre-registered insertion sequence is supplied. For a coupled human–model study, the human may respond adaptively, but those decisions become part of the measured system and should not be described as a controlled common sequence.

9.2.4 Stage four: analyse and preserve

Analysis is performed on both nodes and transitions. Surface text comparison may be included, but the primary questions concern retained distinctions, directional change, uncertainty, provenance, admissibility and later consequences. The full branch histories are preserved so that terminal summaries do not replace the evidence from which they were constructed.

9.3 Suggested experiment manifest

Table 9.1 gives a compact manifest. It can be implemented as JSON, CSV, a database record or a human-readable header, provided the field identities remain stable.

Table 9.1: Suggested trajectory experiment manifest.

Field	Purpose
Parent trajectory ID	Stable identifier for the common external chain.
Parent checksum	Registration that branch copies derive from the same stored bytes or canonical text.
Branch event ID	Identifier shared by all daughter trajectories from one declared split.
Model identity	Provider, displayed model name, version or date where available.
Execution time	Registered start and completion time for each non-simultaneous branch.
Visible start-up condition	System, role, examples, memory and other known framing.
Unknown condition fields	Explicit record of settings or instructions that are not observable.
Insertion sequence	Ordered human, scripted or cross-model perturbations.
Output record	Exact visible output, including interruptions, refusal or tool routing.
Branch policy	Stop after first output, independent continuation, cascade, crossing or synthesis.
Analysis record	Retention, uncertainty, provenance, admissibility and trajectory-deflection measures.
Investigator decisions	Human selections, edits and judgements that altered the later pathway.

9.4 Controls

Three controls are particularly valuable. Repeated runs through the same filter estimate within-model variation. A shuffled or context-reduced parent tests how much the observed continuation depends upon ordering and retained history. A human-coded reference identifies the distinctions whose persistence matters for the specific inquiry. None is a final ground truth. Each provides another finite ruler whose provenance must be retained.

The framework also benefits from negative controls. If two parent chains that should be functionally different produce indistinguishable branch behaviour under the proposed measures, the measurement scheme may be too coarse. If repeated identical presentations appear maximally different, the measures may be dominated by surface wording or uncontrolled stochasticity.

Chapter 10

Measurement Without a Hidden Ideal Answer

10.1 From answer scoring to trajectory comparison

Conventional benchmarks often compare a terminal answer with a stored target. That is useful where the task has an agreed finite outcome, but it does not adequately measure the formation of a conceptual trajectory. In a Functional Symbolic Bifurcation Network, difference is first registered between branches and then interpreted according to the question being investigated.

A general branch difference may be written as

$$D_{AB}(n) = d\left(H_n^{(A)}, H_n^{(B)} \mid C\right), \quad (10.1)$$

where C is the declared comparison condition. The conditioning term matters because no universal semantic distance is assumed. A scientific-argument study may prioritise uncertainty and provenance; a control study may prioritise admissibility and instruction retention; a style study may legitimately prioritise lexical and syntactic structure.

10.2 Possible finite measures

Several measures can be operationalised without pretending that they exhaust meaning. *Distinction retention* records whether specified conceptual separations survive later turns. *Trajectory deflection* records the first node at which a branch changes its explanatory direction or governing vocabulary. *Qualification persistence* records whether conjectural language remains conjectural. *Provenance continuity* records whether sources, origins and earlier decisions remain attached to later claims. *Basin recurrence* records whether a model returns to a characteristic formulation after perturbation.

These may be represented as finite sequences. If c_k is a distinction under study, its retention path in branch j can be written

$$\rho_{j,k} = \left(\rho_{j,k}^{(1)}, \rho_{j,k}^{(2)}, \dots, \rho_{j,k}^{(N)}\right), \quad (10.2)$$

with each value chosen from a declared finite scale such as {retained, transformed, absent, indeterminate}. The sequence shows when a distinction changed, which is more

informative than one average score.

10.3 Compression and symbolic drag

Repeated filtering can compress a trajectory. A long chain of observations, qualifications and provisional names may become a short fluent summary. The summary may be useful, but it can remove the path that made the terminal claims intelligible. Symbolic drag names the residual correction or distortion introduced when the compressed expression is treated as though it carried the whole earlier trajectory.

In a cascade experiment, symbolic drag may be estimated by retaining a set of marked trajectory elements $C = \{c_1, \dots, c_K\}$ and observing their survival after each filter:

$$\sigma_r(c_k) \in \{\text{retained, transformed, lost, indeterminate}\}, \quad (10.3)$$

where r is the cascade position. The objective is not to demand that every detail survive every useful compression. It is to make the removal visible and attributable.

10.4 Uncertainty as part of the measurement

Differences between branch outputs inherit uncertainty from model stochasticity, hidden instructions, version ambiguity, human coding and incomplete access to internal state. This uncertainty should not be discarded when a summary score is produced. It belongs to the nodal and edge records.

An observed difference should therefore be stated conditionally:

$$D_{AB}(n) \mid (P, \delta, C, U), \quad (10.4)$$

where P denotes provenance, δ registered uncertainty, C admissible comparison rules and U explicitly unknown conditions. This follows the wider finite symbolic requirement that a measurement is not separable from the means and history by which it was obtained.

Chapter 11

Connection with the Takens Based Transformer

11.1 A trajectory-sensitive filter

The Takens Based Transformer can be situated within this framework as a deliberately trajectory-sensitive filter. A general LLM may generate a broad candidate continuation, while a smaller local model reconstructs the recent symbolic trajectory, compares it with learned basins and registers whether the candidate is admissible for continuation, routing or storage.

One arrangement is

$$H_n \xrightarrow{\text{general model}} y_{n+1} \xrightarrow{\text{TBT}} \{\text{continue, redirect, hold, reject}\}. \quad (11.1)$$

Another arrangement allows the TBT to act before generation:

$$H_n \xrightarrow{\text{TBT}} c_{n+1} \xrightarrow{\text{general model}} y_{n+1}, \quad (11.2)$$

where c_{n+1} is a finite trajectory constraint. The two arrangements can also be combined to form an entry filter and an exit filter around a general generator.

11.2 Parallel functional channels

The TBT need not collapse all evaluation into one state. Identity, context, uncertainty, admissibility, provenance and task direction can remain as separate functional channels. Each channel reads the same ordered trajectory but reconstructs a different aspect of its continuation. Their outputs can be combined through an explicit nodal value or routing rule.

This is a different use of approximate parallelism. The channels may be computed side by side, but they remain functionally distinct filters over a shared registered chain. A failure in one channel need not be hidden inside a single undifferentiated score.

11.3 Training through neighbouring branches

Functional Symbolic Bifurcation offers a direct method for constructing training basins. A stable parent run-in can be followed by several neighbouring terminal questions or qualifications. Desired continuations can be repeated to deepen one groove, while carefully altered branches teach the model which distinctions should cause a change of direction.

The training object becomes a family of related trajectories rather than a bag of isolated question–answer pairs. This permits deliberate control of neighbouring-path bifurcations. It also offers a way to test compression tolerance: progressively shorten the run-in and observe the point at which the intended response basin can no longer be reconstructed.

11.4 The rotating model architecture

The earlier rotating A–B–C architecture can also be reinterpreted as a controlled trajectory network. An active inference model produces short-term continuations; an updating model incorporates bounded recent trajectories; and a slower consolidator examines a wider historical chain. Their transitions are not merely model replacements. They are provenance-bearing transformations between differently conditioned filters.

The Functional Symbolic Bifurcation Network supplies a way to register what is retained, altered or lost at each rotation. It may therefore help distinguish genuine consolidation from simple rephrasing or catastrophic compression of earlier trajectory distinctions.

Chapter 12

Possible Applications

12.1 Comparative model cartography

Repeated common-parent replay can map characteristic differences among models without reducing them to a ranked leaderboard. Some filters may preserve qualification, some may rapidly normalise unusual terminology, and others may expand one local feature while losing the wider direction. The result is a functional cartography of model response landscapes.

Such maps would be conditional rather than universal. A model's behaviour depends upon the domain, start-up frame, language, context length and date of execution. The cartography would therefore consist of finite maps with stated boundaries, not a final picture of a model's essence.

12.2 Meaning divergence and semantic preservation

A scientific, philosophical or technical formulation can be passed through a model cascade to examine how its distinctions change. This is particularly valuable where fluent paraphrase may conceal conceptual substitution. A trajectory record can show the node at which a tentative proposal became an assertion, a process became a noun, an operational definition became a supposed object, or a provenance-bearing measurement became an unqualified number.

The framework can therefore support empirical study of the Meaning Divergence Crisis. It does not assume that computational and biological meanings must be identical. It records where different filters generate stable shared continuations and where their functional pathways separate.

12.3 Scientific argument and peer review

A manuscript or argument can be represented as a parent trajectory and presented to several reviewing models. Each model's critique becomes a daughter branch rather than a set of interchangeable comments. Follow-up questions can determine whether an objection arises from a local misunderstanding, a stable alternative framework or a difference in the admissible ruler by which the work is assessed.

This is important for work that proposes a new language or framework. A model trained primarily within an established paradigm may repeatedly reconstruct the proposal through the old vocabulary. The branch record can show where incommensurability first appears and whether additional run-in material allows the new distinctions to remain active.

12.4 The held-ruler problem and model self-improvement

A model can evaluate a proposed improvement only through its present filters, language and admissibility rules. If the improvement requires a finer or incommensurable ruler, the current model may be unable to measure it as an improvement. Functional splitting offers a practical response: the existing model and a differently conditioned model can be treated as separate evaluative branches. Neither is silently appointed as the final judge.

The result does not solve the ruler problem, but it makes the transition observable. A new filter can develop a new trajectory of evaluation while the old branch is retained for comparison. Community paradigm change can be represented similarly: the adoption of a new ruler involves the formation of a new network of readers, instruments, terms and admissible transitions.

12.5 Control, routing and tool use

A trajectory-sensitive filter can sit between a broad generator and an exact external interface. It can register task identity, permissions, required fields, uncertainty and provenance before permitting a tool call. Candidate outputs can be routed through specialised filters, with branch histories retained for audit.

This is useful where a single terminal safety score is too coarse. A network can show which constraint caused a route to be accepted, escalated or rejected, and whether the decision remained stable after new information entered the trajectory.

12.6 Model-update monitoring

A stored bank of parent trajectories can be replayed after a provider updates a model. The goal is not merely to see whether answers changed. It is to identify which functional distinctions, uncertainty practices and response basins changed. Because the original parent chains and old branches are retained, the update becomes a provenance-bearing bifurcation across model time.

12.7 Human–AI collaborative inquiry

In ordinary research dialogue, the human and model form a coupled trajectory. A branch can be created deliberately before a major conceptual decision, allowing alternative names, premises or explanatory directions to develop separately. The branches can later be compared or recombined without erasing their different histories.

This may provide a practical research notebook for thought trajectories. Instead of replacing an earlier route whenever a new formulation appears, the investigator can preserve neighbouring paths and return to the exact point at which they separated.

Chapter 13

Discussion

13.1 What the framework provides

The Functional Symbolic Bifurcation Network provides a finite and workable description of multi-turn, multi-model interaction. It replaces the isolated prompt with a provenance-bearing trajectory; replaces the notion of one neutral model operation with a conditioned filter event; replaces classical parallelism with a declared external equivalence and functional split; and replaces terminal answer comparison with the study of daughter trajectories.

The framework is deliberately modest about inaccessible internal states. It does not deny their physical occurrence, nor does it elevate them into an exact semantic reality. It begins with what can be registered: the chain, the branch, the filter identity, the output, the time and the subsequent effect upon continuation. This is sufficient to construct repeatable experiments.

13.2 The strength and danger of the filter metaphor

The filter metaphor is useful because it foregrounds selection, constraint and transformation. Its danger is that it may suggest a pure signal passing unchanged beneath the filters. The present framework does not require such a signal. The parent chain is a finite symbolic registration, not an unmediated meaning. Each model reconstructs a condition from it, and the human reader performs another reconstruction on the resulting output.

There is therefore no need to posit a pristine semantic object behind all branches. Their commonality is established operationally through shared descent from a stored parent. Their differences are then measured according to declared finite rules.

13.3 Network rather than tree

The initial split resembles a tree, but continued interaction produces a network. Outputs can be crossed between branches. A reviewing model can read several daughter trajectories. A human can combine selected portions into a new prompt. A TBT can route a continuation back to an earlier state. These operations create joins, loops and re-entries.

The network representation preserves this complexity without suggesting that all paths occur simultaneously. Every edge remains an ordered registered event. The network is reconstructed from those events after their occurrence.

13.4 The place of classical nonlinear dynamics

Classical nonlinear dynamics may offer useful comparative concepts: attractors, basins, perturbations, hysteresis, recurrence and sensitivity. Each import requires examination. LLM trajectories are finite symbolic sequences; sampling may be stochastic; state dimensions are model-dependent; and the observer often lacks direct access to the supposed phase space. Classical quantities should not be applied merely because the diagrams look similar.

Functional Symbolic Dynamics offers a lower commitment. It permits the finite registration of stable recurrence, trajectory separation and regime change before deciding whether a classical mathematical structure is admissible. Where a classical model later predicts or organises the observations successfully, it can be retained with its conditions and uncertainty.

13.5 The human filter cannot be removed

The human investigator chooses the parent chain, the branch point, the models, the follow-up insertions and the measures of difference. Even an automated experiment inherits those choices in code and dataset construction. The human filter is therefore part of the provenance.

This is not merely contamination to be eliminated. In collaborative inquiry, the human capacity to recognise an unexpected distinction may be the reason for performing the experiment. The requirement is to record where judgement enters, distinguish scripted from adaptive continuation, and avoid presenting a human-selected path as though it emerged independently from the machines.

Chapter 14

Open Questions

14.1 What constitutes the same parent?

Is byte identity required, or is a normalised text sufficient? Should timestamps, file attachments, formatting and interface metadata form part of the parent? A strict answer will depend upon the experiment. The framework needs a hierarchy of external equivalence relations, ranging from exact stored bytes to declared semantic or functional equivalence. Each relation supplies a different ruler and must be named.

14.2 Where does a branch begin?

The visible branch point may be the moment at which the same transcript is sent to another model. Yet divergence may already have entered through hidden wrappers, model updates or prior memory. A branch therefore begins operationally at the earliest registered split, not necessarily at the first physical difference. How unknown preconditions should be represented remains an important methodological question.

14.3 How should semantic distance be measured?

Vector similarity is convenient but may miss changes in qualification, provenance or causal direction. Human coding is sensitive but introduces another filter. A TBT or other trajectory-sensitive model may offer an intermediate method, but it too requires training and validation. A plural measurement system may be preferable to one universal semantic metric.

14.4 How long must a branch continue?

A one-turn difference may disappear immediately; a subtle distinction may affect an answer only many turns later. The observation window must be long enough to test persistence, but long branches also accumulate uncontrolled differences. Methods are needed for selecting branch depth and for distinguishing early cause from later amplification.

14.5 What is a stable response basin?

Repeated terminology is not sufficient. A model may use similar words while changing their function. Conversely, it may preserve a functional distinction while changing vocabulary. Basin identity may need to be defined through the set of admissible later continuations rather than surface form. Constructing a finite test of that set is an open problem.

14.6 How does context compression alter the network?

Context windows are finite. Earlier messages may be truncated, summarised or internally compressed. The effective history presented to the model can therefore differ from the stored transcript. The node record should distinguish the held external history from the history actually available to the filter where this can be determined. Compression boundaries may themselves cause behavioural regime transitions.

14.7 Can branches genuinely reconverge?

Two outputs may become lexically similar while retaining different provenance and later potential. Is reconvergence defined by visible text, nodal values, future response behaviour or human judgement? The framework needs a criterion that does not erase history merely because terminal states resemble one another.

14.8 How should unknown filters be represented?

Commercial systems may apply undisclosed system messages, routing models, retrieval, safety layers and model substitutions. These cannot simply be ignored, but nor can they be specified without evidence. One option is to represent the service as a compound black-box filter with a time-stamped identity and an explicit unknown-condition vector. Repeated control trajectories may then reveal changes without assigning an unsupported internal cause.

14.9 What is the role of registered time?

Branches executed at different moments are only approximately parallel. Model deployments can change between runs, and the human's own understanding can change during the experiment. Registered time, inferred model time, symbolic trajectory position and relational human-model time may need separate treatment. A fuller temporal model could connect this network with the Temporal Reference Manifold.

14.10 Can the framework guide new model construction?

The present framework can analyse existing systems, but it may also suggest architectures that preserve branch identity, provenance and uncertainty directly. A model could retain alternative trajectories instead of collapsing immediately to one continuation; route different

functional channels through specialised filters; or use explicit reconvergence rules. Whether these structures improve practical reasoning remains an experimental question.

Chapter 15

Conclusion

An AI interaction can be treated as a finite nonlinear dynamical process without claiming access to an invisible continuous semantic world. It begins from a start-up network condition, proceeds through an ordered train of history-conditioned filter events, and returns each visible output to the symbolic record from which later states are reconstructed. The conversation therefore has memory, feedback, path dependence and the capacity for divergence.

When a held chain is presented to several models, the operation is not classically parallel. It is an externally registered equivalence followed by a functional split. Each model repartitions the chain and reconstructs a different operative condition. The returned outputs become daughter trajectories. Their later development can be held, compared, continued, crossed, cascaded or recombined.

Functional Symbolic Dynamics provides the appropriate level of description because it preserves finite states, ordering, provenance, uncertainty and admissibility. It allows a Functional Symbolic Bifurcation Network to be constructed from events that can actually be performed and recorded. The framework does not require all branch differences to be called classical bifurcations. It separates deliberate architectural splitting from persistent response-regime change and supplies experiments through which the latter may be investigated.

Perhaps the most important consequence is methodological. The output of a model need not be treated as an isolated answer, and disagreement among models need not be reduced to competition for one hidden ideal response. Each answer is a registered continuation produced by a particular filter acting upon a particular history. By retaining the branch and the path that generated it, we gain a practical way to study how meanings, uncertainties, assumptions and conceptual rulers develop across human and artificial systems.

The proposed network is therefore both an analytical object and a possible machine architecture. It can map existing interactions, support trajectory-sensitive control, guide TBT training, preserve alternative routes of inquiry and make visible the transformations introduced by repeated model filtering. Its open questions are not weaknesses to be concealed. They identify the measurements, experiments and language that must now be developed if the framework is to become a mature branch of Functional Symbolic Dynamics.

Appendix A

Notation

Symbol	Meaning
H_n	Held externally registered trajectory after position n .
u_n	Human, scripted or cross-model insertion at position n .
$y_n^{(j)}$	Visible output registered from model-filter j .
$S_0^{(j)}$	Start-up network condition for model-filter j .
W_j	Model-specific interface wrapping or presentation operation.
R_j	Reconstruction of an operative internal condition.
F_j	Nested model-conditioned filter.
P_j	Projection into a finite visible response.
B_n	Registered Functional Symbolic Bifurcation event.
$N_{j,n}$	Dynamic Semantic Lattice node for filter j at trajectory position n .
$\nu(N_{j,n})$	Finite Nodal Value record.
$\sim_{\text{ext}}$	Declared external symbolic equivalence, not endogenous identity.
$\parallel$	Finite ordered concatenation.

Appendix B

Minimal Branch Record

The following plain-text form can accompany a branch experiment:

```
EXPERIMENT_ID:
PARENT_TRAJECTORY_ID:
PARENT_CHECKSUM:
BRANCH_EVENT_ID:

MODEL_DISPLAY_NAME:
MODEL_VERSION_OR_DATE:
INTERFACE:
EXECUTION_START:
EXECUTION_END:

VISIBLE_SYSTEM_OR_ROLE_CONDITION:
KNOWN_MEMORY_CONDITION:
KNOWN_DECODING_CONDITION:
UNKNOWN_CONDITIONS:

PARENT_TRAJECTORY:
BRANCH_INSERTION:
MODEL_OUTPUT:

BRANCH_POLICY:
FOLLOW_UP_SEQUENCE:
HUMAN_INTERVENTIONS:

DISTINCTIONS_TO_TRACK:
UNCERTAINTY_RECORD:
PROVENANCE_NOTES:
ANALYSIS_NOTES:
```

Appendix C

Provisional Definitions

Functional Symbolic Trajectory. A finite, ordered and provenance-bearing pathway of symbolic states and registered transformations through which later symbolic function is conditioned.

Model-conditioned filter. A finite operative composition through which a presented symbolic trajectory is partitioned, reconstructed, constrained and projected as a visible continuation.

Start-up network condition. The inherited and inserted condition from which the first registered continuation in an experiment is produced, including known model, instruction, interface, memory and decoding conditions and an explicit record of what remains unknown.

Functional Symbolic Bifurcation. The registered copying, routing or re-presentation of one held finite symbolic trajectory to two or more separately conditioned filters, producing daughter continuations with preserved common provenance.

Functional Symbolic Bifurcation Network. A finite directed network whose nodes are registered trajectory states, whose edges are observed filter-conditioned transformations, and whose branch, crossing and convergence events retain provenance, uncertainty and admissibility.

Approximate parallelism. A declared relation among separately executed branches that share an externally equivalent parent trajectory while differing in physical time, model condition or endogenous reconstruction.

Behavioural regime transition. A persistent and reproducible alteration in the functional pattern of continuation following a finite symbolic or model-condition perturbation. It is not assumed to be identical to a classical continuous-system bifurcation.

Closing Note

This monograph establishes a working foundation rather than a closed theory. Its diagrams and notation are intended to be revised as practical experiments reveal which distinctions are productive. The essential commitment is that the parent chain, its split, its filters and its daughter histories remain finite, registered and available for examination. In that retained trajectory lies the possibility of moving from an impression that models “answer differently” to a measured account of how symbolic pathways are reconstructed, redirected and stabilised.